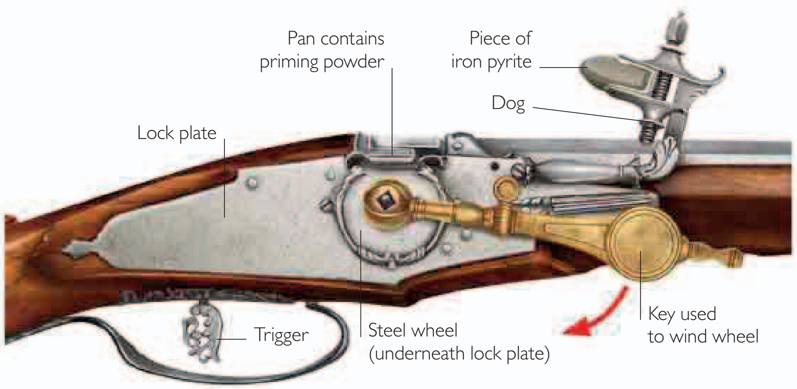
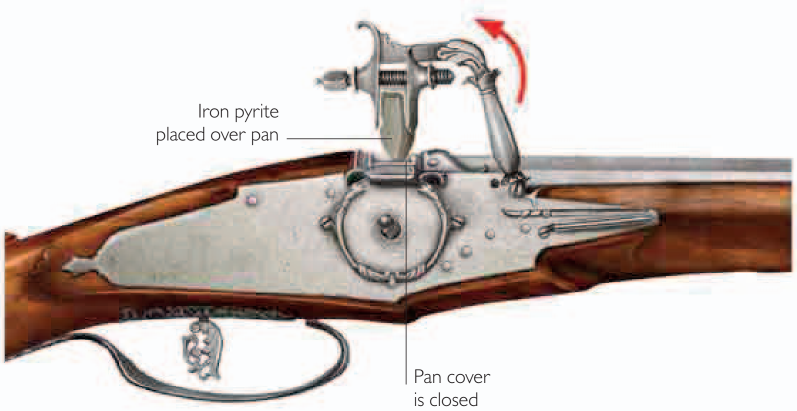
Wheel-lock

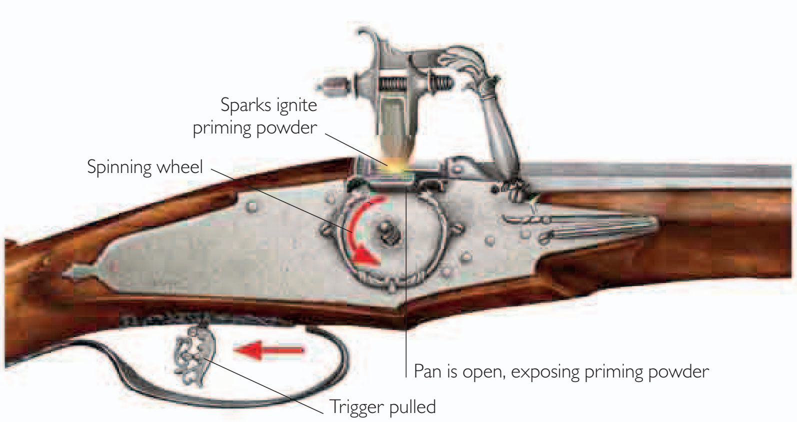
The wheel-lock used a rotating steel wheel to strike sparks from a piece of iron pyrite. After loading the barrel, the user rotated the wheel with a key about three-quarters of a turn, until it was held by the trigger mechanism. Then he placed the priming powder in the pan. The top of the wheel passed up through a slot in the bottom of the priming pan, so that sparks produced when the iron pyrite contacted the wheel fell into the priming powder.



1 A spring-loaded arm called a dog, retained in position by the dog spring, holds a piece of iron pyrite in its jaws. The user spans the lock—winding the steel wheel using a key, which compresses the mainspring (underneath lock plate).



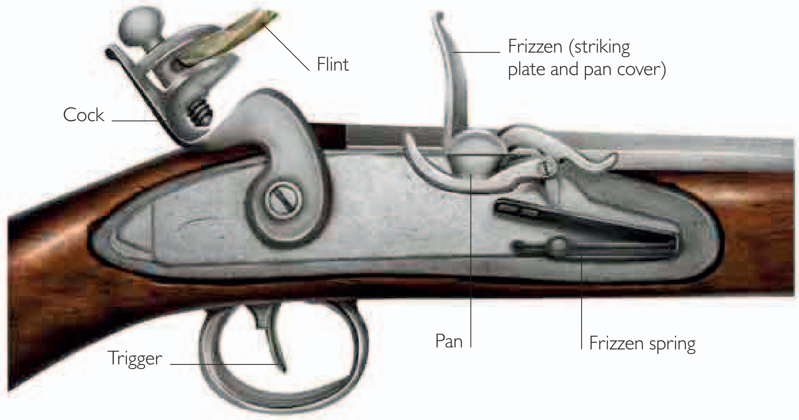
2 Before firing, the user moves the dog manually, placing it onto the pan cover, which is shut.



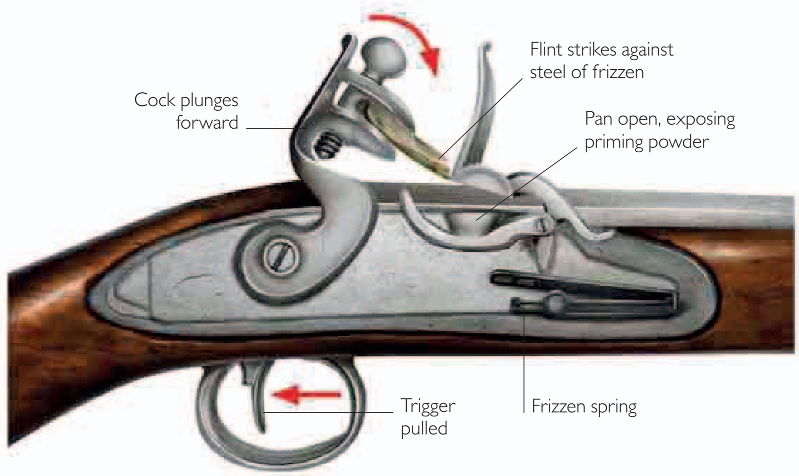
3 Pulling the trigger releases the wheel, which starts spinning. The pan cover opens automatically, bringing the iron pyrite into contact with the wheel. The friction creates sparks, which ignite the priming powder, causing a flash that ignites the main charge in the barrel.

Flintlock

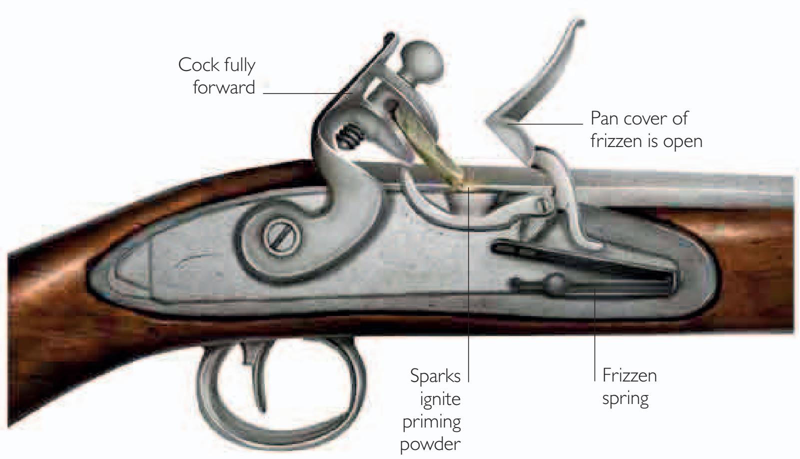
The flintlock had a simpler design than the wheel-lock. It used the impact of natural flint on hardened steel to strike sparks. The cock held a flint, which was propelled forward by a spring to strike a steel part called the frizzen, which was a combined striking plate and pan cover. The impact forced the steel back, opening the pan cover. Sparks fell into the priming powder to ignite it.



1 Before firing, the cock is held by a hooked part called a sear (inside the gun). A frizzen spring holds the frizzen closed over the pan.



2 Pulling the trigger retracts the sear, allowing the cock to spring forward to scrape the face of the steel. This impact forces the steel back, opening the attached pan cover and exposing the priming powder.



3 Sparks caused by the flint striking the steel fall into the pan to ignite the priming powder. This produces a flash that ignites the main charge in the barrel via a vent in the side of the barrel.